

CODE SPORT



In this month's column, we return to our discussion on natural language processing.

In the last two columns, we have been discussing computer science interview questions. This month, let's continue our ongoing discussion on natural language processing (NLP) by looking at one of its emerging applications.

Automation is one of the areas likely to benefit from NLP techniques and tools. Many of us have interacted with customer service agents from different sectors such as e-commerce, banking, hospitality, etc, over the phone, via email or chat, at all hours of the day, 24x7. Some of these communication channels are in real-time, such as the phone or chat, while others are offline, such as email.

Today, while these services are provided by human agents, there is an increasing push towards automation. The chat typically starts off with the agents greeting the customer and asking what they can do for the latter. The customers then state their problem or put forth their query, which is then followed by the agents asking for more details from the customers to authenticate their identity, obtain the order information, and then provide steps to answer the query or solve the problem. This is an interactive conversation which has a typical structure and follows a well-defined pattern of dialogue between the customer and agent. The agents are typically provided with a 'FAQ' or 'problem-solution' manual in which information on resolving customer queries is systematically documented, with details on each step the agent needs to follow. The conversations are audited to ensure that the agents were courteous in their communication with customers, that they follow the appropriate problem-resolution script, and that they use the appropriate greetings at the start and end of communication.

Let us consider a simple scenario. You have ordered an item on an e-commerce website and have not yet received it. So you are using the Web chat interface to speak to the customer service agent. After the customary greeting, you tell the agent that you have ordered item XYZ one week back but it has not yet been delivered.

The agent then asks you for the order details. He then apologises to you for putting you on hold while he looks up the backend database. Once he has the details, he either mentions that, "The item has already been shipped and you can expect to receive it on date XX-YY-ZZ," or he tells you that the delivery has been delayed and gives you the reason for the delay, etc. Now, if you analyse the many thousands of such conversations that take place every day, you can find that they follow a typical pattern. From that, the crucial next question that springs to mind is, "Can these queries be answered by an automated agent instead of a human agent?"

Well, the answer is yes, and NLP comes to our rescue in automating these communication services. While at first glance it seems quite difficult to replace the human-to-human communication with human-to-virtual agent communication, it is indeed possible. In many cases, virtual agents provide initial communication support for simple queries. However, human agents can take over in case the conversation becomes complex and deviates a lot from the normal script.

There are two key components to a solution to virtualise the communication between human customer and virtual agent. First, the human customer's part of the conversation needs to be analysed and then the appropriate response and follow-up question, if any, needs to be generated in natural language format and fed to the virtual agent. So both natural language processing and natural language generation need to be done. The key challenge is to ensure that the conversational experience does not deteriorate by automating the process and the human customer does not feel dissatisfied (of course, the ideal experience would be when the human customer cannot detect whether he is talking to a virtual agent or a human agent).

Let us analyse the problem in a little more detail, focusing first on the natural language processing part. Let us assume that we are given a database of thousands of chats that have happened between human agents and